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CLASS: BSAI **SECTION:3A**

LAB TASK: 1

TASK:1

History of AI

Below is a table listing some significant events throughout the development of AI:

Alan Turing proposed the Turing Test (Imitation Game)

This was Alan Turing asking if you could design a machine that could imitate human behaviour to such an extent that it would be impossible to distinguish between the two. This provided a platform for people to discuss the intelligence of machines.

Mind, Turing, A.M. (1950)

1956

Dartmouth Summer Research Project on Artificial Intelligence

Many researchers came together to investigate whether machines can learn and think. The term 'Artificial Intelligence' was used in this project and AI became seen as a separate area of study.

McCarthy et al (1955) Dartmouth proposal

1966

ELIZA Chat Bot – Joseph Weizenbaum

ELIZA would look for key words in the users message and respond based on pre-defined templates. This was one of the first programs for chatting with a computer using natural language.

Weizenbaum, J. (1966) Communications of the ACM

1974-1980

First AI Winter

Initial AI had many promises but didn't perform well. This caused people to lose interest in AI. Funding and support was reduced.

(Lighthill Report, 1973)

1997

IBM Deep Blue beats world chess champion (Garry Kasparove)

This showed that with fast hardware and well designed searching algorithms computers can outperform humans at difficult tasks.

Campbell, Hoane and Hsu (2002) Artificial Intelligence journal

2012

Alex Net wins ImageNet competition

Alex Net was a deep convolutional neural network trained using GPUs. It achieved much better results than any other entries.

TASK:2

Program to determine pass/fail

```
def check_result(score, documents, attendance, verification):

if score >= 70 and documents == True and attendance >= 75 and verification >= 60:

return "Pass"

else:

return "Fail"

test_cases = [ (85, True, 80, 75, "Pass", "All conditions are met"), (85, True, 70, 75, "Fail", "Attendance is less than 75"), (85, True, 80, 55, "Fail", "Verification score is less than 60"), (70, True, 75, 60, "Pass", "All values are exactly at the limit")]

number = 1

for case in test_cases:

score = case[0]

documents = case[1]

attendance = case[2]

verification = case[3]

expected = case[4]

reason = case[5]

actual = check_result(score, documents, attendance, verification)

print(f"Test Case {number}")

print(f"Input: {(score, documents, attendance, verification)}")

print(f"Expected: {expected}")

print(f"Actual: {actual}")

print(f"Reason: {reason}")
```

print("----------")

number += 1

Test Case 1

Input: (85, True, 80, 75)

Expected: Pass

Actual: Pass

Reason: All conditions are met.

Test Case 2

Input: (85, True, 70, 75)

Expected: Fail

Actual: Fail

Reason: Attendance is less than 75.

Test Case 3

Input: (85, True, 80, 55)

Expected: Fail

Actual: Fail

Reason: Verification score is less than 60.

Test Case 4

Input: (70, True, 75, 60)

Expected: Pass

Actual: Pass

Reason: All values are exactly at the limit.

Results

Test Case

Input

Expected

Actual

Reason

1

(85, True, 80, 75)

Pass

Pass

All conditions are met.

2

(85, True, 70, 75)

Fail

Fail

Attendance is less than 75.

3

(85, True, 80, 55)

Fail

Fail

Verification score is less than 60.

4

(70, True, 75, 60)

Pass

Pass

All values are exactly at the limit.

TASK: 3

Chat test and General Intelligence

Imagine an AI that chats with students that want to apply for a scholarship. If an AI can answer questions, tell them if they are eligible and explain what they need to do to apply it has passed a chat test. However, just because an AI can chat naturally doesn't mean that it has general intelligence.

In 1950 Alan Turing proposed the imitation game. This was a way to discuss machine intelligence by having a chat. However, this only shows that the AI can reply well. It doesn't show that the AI actually understands the rules, investigates facts and makes

correct decisions. This is similar to Searle (1980) Chinese Room argument, where he says a program can follow rules to provide answers.

False logic

If a student asks:

‘Can I get the scholarship if I have 66 marks? The minimum is 70.’

The AI may reply:

‘Yes! You have done well and your documents show you are a hardworking student.’

This seems friendly and natural but is incorrect as 66 is less than 70. The student would be confused.

Better test:

A better test would be to give the AI 4 applications for scholarships. Each application will have different scores and documents.

Conclusion

A chat test will only show that the AI can chat naturally. Whereas a practical test will show that the AI can do the job correctly. Therefore, passing a chat test cannot be used to determine general intelligence.

GITHUB LINK:

https://github.com/umair-900/AI-LAB-TASKS
